15-WEEK COMPLETE PROGRAM

FULL STACK WEB DEVELOPMENT

HTML · CSS · Bootstrap · JavaScript · React · SQL · Node.js · MongoDB · AWS · Gen AI — from zero to deploying real full-stack applications.

15 Weeks	11 Modules	5+ Projects	MERN Stack	100% Support
Duration	Curriculum	Hands-On	Tech Focus	Placement

YOUR LEARNING JOURNEY

A logical, step-by-step path — starting from zero, ending with full production-ready apps.

W0 **Dev Setup** GitHub, Notion, VS Code	W1 **HTML** Structure & semantics	W2 **CSS** Styling & Flexbox	W3 **Bootstrap** Responsive UI	W4–5 **JavaScript** Core JS + ES6+ + DOM	W6–7 **React.js** Components, Hooks, Router
W8 **SQL** Databases & Queries	W9–10 **Node.js** Server & REST APIs	W11–12 **MongoDB** NoSQL + Mongoose	W13 **Deployment** AWS EC2 + CI/CD	W14–15 **Advanced** Redis, Payments, AI	■ **Gen AI** ChatGPT API in Apps

MODULE BREAKDOWN

Every key concept covered — concise, complete, and industry-relevant.

MOD
00
WEEK 0

Dev Setup & Tools

- GitHub — version control basics
- Notion — project management
- Slack — team communication
- VS Code setup & extensions
- Terminal basics & shortcuts
- Chrome DevTools overview

■ OUTCOME ***Be fully set up like a professional developer from Day 1***

MOD 01 — WEEK 1

HTML — Web Structure

- HTML Boilerplate & document structure
- Headings, Paragraphs, Lists, Links
- Images, Buttons & Anchor Tags
- Tables: TR, TD, TH elements
- Forms: Input types, Labels, Validation
- Semantic HTML5 elements

OUTCOME ***Build any web page structure from scratch with proper semantic markup***

MOD 02 — WEEK 2

CSS — Styling & Layouts

- Selectors, Properties & Specificity
- Color systems: RGB, Hex, HSL
- Box Model: Margin, Padding, Border
- CSS Flexbox — full layout control
- Font Family, Size & Text properties
- Responsive units: px, %, rem, vh/vw

OUTCOME ***Style beautiful, responsive web pages and master Flexbox layouts***

MOD 03 — WEEK 3

Bootstrap — Rapid UI

- Bootstrap Grid system (12-col responsive)
- Buttons, Badges, Alerts, Cards
- Typography & Utility classes
- Navbars, Modals & Forms
- Bootstrap Icons library
- Custom theming & overrides

OUTCOME ***Build polished, mobile-first UIs in a fraction of the time***

MOD 04 — WEEK 4–5

JavaScript — Core to ES6+

- Variables: var, let, const; Data Types
- Operators, Conditionals & Loops
- Functions, Scope & Arrow Functions
- Arrays & Objects — full manipulation
- DOM manipulation & Event Listeners
- Async JS: Promises, async/await, Fetch API
- ES6+: Destructuring, Spread, Modules
- OOP: Classes, Inheritance, Prototypes

OUTCOME ***Write dynamic, interactive web applications with modern JavaScript***

MOD 05 — WEEK 6–7

React.js — Frontend Framework

- Components: Functional & Class
- JSX syntax & rendering logic
- Props & State management
- React Hooks: useState, useEffect, useRef
- React Router — SPA navigation
- Context API & lifting state
- Forms & controlled components
- Connecting to APIs with Axios / Fetch

OUTCOME ***Build fast, component-driven single-page applications with React***

MOD 06 — WEEK 8

SQL & Databases

- Relational DB concepts — tables, keys
- DDL: CREATE, ALTER, DROP
- DML: SELECT, INSERT, UPDATE, DELETE
- JOINs: INNER, LEFT, RIGHT, FULL
- Aggregate functions & GROUP BY
- Subqueries, CTEs & Window Functions
- Indexing & query optimisation
- User auth & DB security basics

OUTCOME ***Design and query relational databases for any backend application***

MOD 07 — WEEK 9–10

Node.js & Express — Backend

- Node.js architecture & event loop
- Modules: require, module.exports
- NPM, package.json & dependencies
- Express framework & middleware
- REST API design — GET/POST/PUT/DELETE
- Path params, query strings & req body
- JWT Authentication & authorisation
- Security: bcrypt, helmet, CORS

OUTCOME ***Build secure, scalable RESTful APIs and backend servers***

MOD 08 — WEEK 11–12

MongoDB & Mongoose

- NoSQL vs SQL — when to use what
- Installing & connecting MongoDB
- CRUD: insertMany, find, updateOne, deleteOne
- Mongoose schemas & models
- Schema validation & constraints
- Mongoose middleware & virtuals
- Aggregation pipeline
- Connecting MongoDB + Node.js + Express

OUTCOME ***Design flexible NoSQL databases and connect them to your Express backend***

MOD

09

WEEK 13–15

Deployment, Projects & Advanced

- Full-stack MERN capstone project
- REST API integration end-to-end
- AWS EC2 — deploying Node.js apps
- CI/CD basics & environment variables
- Redis — caching for performance
- Microservices vs MVC architecture
- Multer — file uploads
- Payment Gateways & Email / Notification APIs

OUTCOME ***Deploy real production applications and build an impressive portfolio***

MOD

BONUS

Generative AI — Build Smarter Web Apps

- ChatGPT API — OpenAI integration in Node.js
- Prompt engineering for developers
- Build an AI chatbot feature in your app
- GitHub Copilot — AI pair programming
- AI-powered code review & debugging
- Gemini API & multi-model usage
- LangChain basics — chaining AI workflows
- AI for content generation in React apps

OUTCOME ***Integrate AI features into your full-stack apps and 10x your productivity***

TECH STACK YOU WILL MASTER

15 industry-standard tools covered across the full program.

HTML5 Markup	**CSS3** Styling	**Bootstrap** UI Framework	**JavaScript** Language	**React.js** Frontend
Node.js Runtime	**Express.js** Backend	**MongoDB** Database	**SQL / MySQL** Relational DB	**REST APIs** Integration
GitHub Version Control	**VS Code** IDE	**AWS EC2** Deployment	**Redis** Caching	**ChatGPT API** Gen AI

WHO IS THIS FOR?

- **Fresh Graduates**
  Any degree — BCA, B.Tech, BSc, Arts, Commerce
- **Career Switchers**
  Zero coding background? Start right here
- **Working Professionals**
  Upskill and move into developer roles
- **Entrepreneurs**
  Build your own web products & MVPs
- **Designers**
  Add coding skills to your design toolkit

WHY CODING SHARKS?

- **Industry Expert Mentors**
  Sessions led by working software engineers
- **Project-Based Learning**
  5+ real full-stack projects for your portfolio
- **Full Placement Support**
  Resume, LinkedIn, mock interviews, campus drives
- **Industry Certificate**
  Recognised completion certificate on graduation

Start Building Real Apps Today!

Call / WhatsApp
+91 77470 04451

Visit :
www.thecodingsharks.in